Modelling of Advective-Diffusive-Reactive Flows in Geological Carbon Sequestration

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Declaration

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Abstract

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Acknowledgements

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1 Introduction

The solubility trapping of geologically sequestered CO_2 represents one of the most secure, long-term solutions to underground geological carbon storage - second only to the much slower process of mineral trapping - because unlike CO_2 in liquid phase, brine containing dissolved CO_2 becomes denser and thus does not present any risk of rising by buoyancy forces to the top of the geological formation and escaping through structural faults in the cap rock. Convective dynamics driven by such density contrasts have been well-studied in the literature on porous media flows [**?**, **?**]. However, the solubility of CO_2 in water is relatively low and the timescale for the onset of convection processes is estimated to be large [**?**]. One tier down from solubility trapping in the hierarchy of storage security [**?**] (see Figure 1.1) is capillary trapping, whereby small amounts of CO_2 are left behind in the wake of a migrating CO_2 current. The interplay between capillary and solubility trapping provides the motivation for the research forming this thesis. As small bubbles of CO_2 form and are left trapped in pore spaces, nearby brine is expected to partially dissolve these bubbles and become saturated. If the saturated brine is then replaced by unsaturated brine, due to mechanisms such as convection, exchange flow or background hydrogeologic flow, the capillary-trapped CO_2 would further diminish by dissolution, and potentially dissolve entirely.

1.1 One

1.2 Two

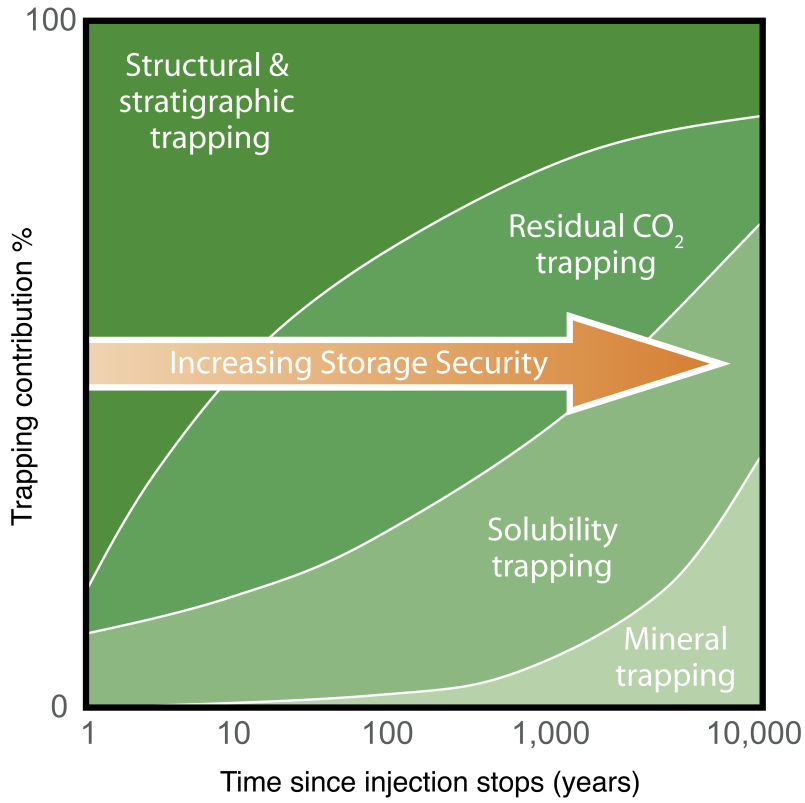

Figure 1.1: Mechanisms of geological CO_2 storage, reproduced from (**metz_ipcc_2005?**).

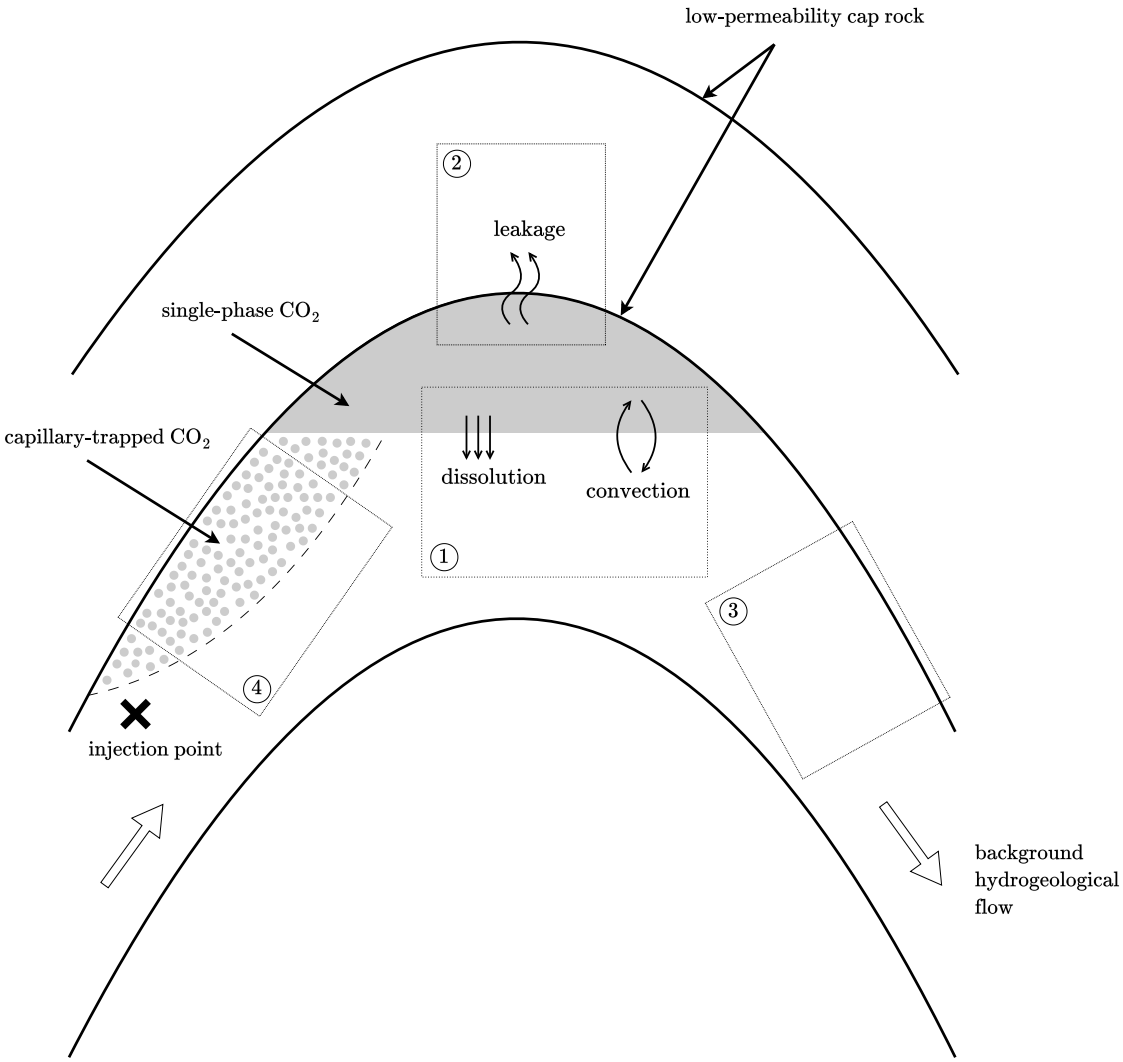

Figure 1.2: Schematic of geologically-sequestered CO_2 in an anticline formation. Single-phase CO_2 is injected at a point downslope of the apex, but rises towards the apex because it is more buoyant than the surrounding brine. Capillary-tragged CO_2 is left behind in its wake. Further rising of the buoyant CO_2 is halted by the impermeable cap rock layer, though leakage may still occur through faults, or over a much longer time scale if the cap rock has low but non-zero permeability. Some anticline formations may also have a background hydrogeological flow from on side to the other. Boxes 1, 2 and 3 isolate the different regions and flows within the anticline formations that research into the fluid dynamics of carbon sequestration has investigated so far with simplified mathematical and numerical models. A simplified model for box 4 is proposed in the following section.

2 Governing equations for the dissolution of capillary-trapped CO_2

2.1 Porosity & saturation

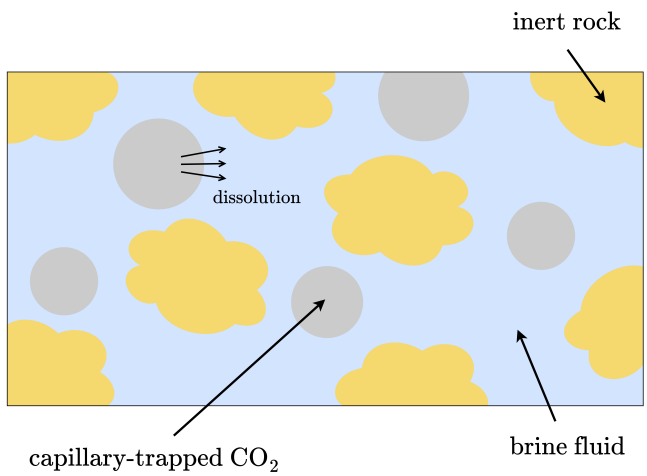

Figure 2.1: Pore-scale schematic illustrating the partition of space into brine fluid, capillary-trapped CO_2 and rock. 'Bubbles' or ganglia of CO_2 are held in place by surface tension forces, but dissolve into the surrounding brine. The rock is assumed inert, reacting with neither CO_2 nor brine.

We consider a porous medium consisting of rock, brine fluid and capillary-trapped CO_2 as illustrated in Figure 2.1 , such that the partition of space can be expressed as

$$\phi + \phi_{\mathrm{CO_2}} + \phi_{\mathrm{rock}} = 1 \tag{2.1}$$

where the porosity ϕ is the volume fraction of brine fluid and the remaining volume fractions $\phi_{\mathrm{CO_2}}$ and ϕ_{rock} are contributed by the capillary-trapped CO_2 and rock, respectively. As modelling assumptions we assume that the rock is inert and the 'bubbles' of capillary-trapped CO_2 held in place by surface tension forces are immobile, so that only the brine fluid is

mobile. However the amount capillary-trapped CO_2 is able to gradually diminish over time by dissolution into the neighbouring brine. In general, brine and single-phase CO_2 are a pair of immiscible liquids that can be modelled as a multiphase flow, but in this thesis we only investigate the considerably simpler problem of the single-phase flow of brine fluid containing dissolved CO_2. Given that we are primarily interested in the spatial extent and temporal evolution of the capillary-trapped CO_2, we can write the effective porosity

$$\phi(\mathbf{x}, t) = \varphi(\mathbf{x})(1 - s(\mathbf{x}, t)) \tag{2.2}$$

in terms of the saturation $s(\mathbf{x}, t)$ of capillary-trapped CO_2 and the porosity distribution $\varphi(\mathbf{x})$ of the inert rock formation, which is in general heterogeneous. Since the rock is by assumption inert, $\frac{\partial \varphi}{\partial t} = 0$ and any evolution in the effective porosity is solely due to evolution in the saturation. $s = 0 \Leftrightarrow \phi = \varphi$ corresponds to the pore space containing only brine fluid CO_2, whereas at the other extreme $s = 1 \Leftrightarrow \phi = 0$ corresponds to pore space containing only capillary-trapped CO_2. In term of s and φ, the the other volume fractions are $\phi_{CO_2}(\mathbf{x}, t) = \varphi(\mathbf{x})s(\mathbf{x}, t)$ and $\phi_{\text{rock}}(\mathbf{x}) = 1 - \varphi(\mathbf{x})$.

2.2 Governing equations

In a porous medium filling a domain Ω with boundary $\partial\Omega$, flow coupled to thermosolutal transport is modelled by the general governing equations for $(\mathbf{x}, t) \in \Omega \times [0, \infty)$

$$\phi \frac{\partial \theta}{\partial t} + \nabla \cdot (\theta \mathbf{u} - \mathsf{G} \cdot \nabla c) = 0 \tag{2.3}$$

$$\frac{\partial (\phi c)}{\partial t} + \nabla \cdot (c\mathbf{u} - \mathsf{D} \cdot \nabla c) = R \tag{2.4}$$

$$\frac{\partial (\phi \rho)}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = R \tag{2.5}$$

$$\mathbf{u} = -\frac{\mathsf{K}}{\mu} \cdot (\nabla p - \rho \mathbf{g}) \tag{2.6}$$

$$\varrho \frac{\partial \phi}{\partial t} = R \tag{2.7}$$

where the transport equation (2.3) for the fluid temperature θ with a thermal dispersion G is derived from the conservation of energy (assuming no heat transfer between the solid and liquid phases in the porous medium and no heat sources or sinks); (2.4) describes the transport of solute dissolved at concentration c (solute mass per unit fluid volume) in fluid flowing through a porous medium of porosity ϕ, incorporating advective transport due to the Darcy velocity $\mathbf{u}$, dispersion with coefficient D and rate of addition or removal R due to chemical reactions;

(2.5) expresses mass conservation in the fluid of density ρ; (2.6) is Darcy's law expressing the Darcy velocity $\mathbf{u}$ in terms of the permeability K, fluid viscosity μ, pressure gradient ∇p and buoyancy force $\rho\mathbf{g}$; (2.7) also expresses mass conservation and models the evolution of porosity ϕ due to the addition or removal of solute mass from an immobile phase of density ϱ. Note that the Darcy velocity differs from the interstitial fluid velocity $\phi\mathbf{u}$ by a factor of porosity. This system of equations is closed by a set empirical relations for the concentration-dependent fluid density $\rho = \rho(c)$ and viscosity $\mu = \mu(c)$ (i.e. equations of state), porosity-dependent permeability $\mathsf{K} = \mathsf{K}(\phi)$, dispersions $\mathsf{G} = \mathsf{G}(\phi, \mathbf{u})$ and $\mathsf{D} = \mathsf{D}(\phi, \mathbf{u})$ dependent on both porosity and Darcy velocity, and a reaction rate $R = R(\phi, c, \theta)$ dependent on porosity, concentration and temperature. G, D and K are in general tensor-valued to account for anisotropy in the porous medium. The gravity vector expressed in terms of its unit vector is $\mathbf{g} = g\mathbf{e}_g$.

The governing equations (2.3)-(2.7) are supplemented by a set of boundary conditions on θ, c, and ($\mathbf{u}$, p) and initial conditions on θ, c and s. In general the boundary conditions on the transport equations (2.3) and (2.4) may be a combination of Dirichlet (D) and Neumann (N) boundary conditions prescribing a known value or flux respectively

$$\theta = \theta_{\mathrm{D}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{D},\theta} \tag{2.8}$$

$$\mathbf{n} \cdot (\mathsf{G} \cdot \nabla\theta) = \theta_{\mathrm{N}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{N},\theta} = \partial\Omega / \partial\Omega_{\mathrm{D},\theta} \tag{2.9}$$

$$c = c_{\mathrm{D}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{D},c} \tag{2.10}$$

$$\mathbf{n} \cdot (\mathsf{D} \cdot \nabla c) = c_{\mathrm{N}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{N},c} = \partial\Omega / \partial\Omega_{\mathrm{D},c} \tag{2.11}$$

where $\mathbf{n}$ is the outward unit normal to the boundary. The boundary conditions on the fluid flow governed by the Darcy system (2.5) and (2.6) may be either a prescribed pressure, or a prescribed normal velocity. For reasons that will become evident later in our discussion of the finite element method, we shall refer to these as essential (E) and natural (N) boundary conditions respectively.

$$\mathbf{n} \cdot \mathbf{u} = u_{\mathrm{E}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{E}} \tag{2.12}$$

$$p = p_{\mathrm{N}} \quad \text{for } \mathbf{x} \in \partial\Omega_{\mathrm{N}} = \partial\Omega / \partial\Omega_{\mathrm{E}} \tag{2.13}$$

In the case $\partial\Omega_{\mathrm{N}} = \emptyset$, the velocity boundary condition must also obey the constraint $\int_{\partial\Omega_{\mathrm{E}}} u_{\mathrm{E}} \, \mathrm{d}\Gamma = 0$ due to incompressibility. Since the governing equations do not contain any time-derivatives of the velocity or presure, initial conditions on $\mathbf{u}$ and p need not be provided. We denote the initial temperature, concentration an saturation by

$$\theta(\mathbf{x}, t = 0) = \theta_0(\mathbf{x}) \tag{2.14}$$

$$c(\mathbf{x}, t = 0) = c_0(\mathbf{x}) \tag{2.15}$$

$$s(\mathbf{x}, t = 0) = s_0(\mathbf{x}) \tag{2.16}$$

In our model of the dissolution of capillary-trapped CO_2 and its subsequent transport via advection and diffusion in the brine fluid, c is the concentration of CO_2 dissolved in brine, R is the rate of addition into the brine due to the dissolution of capillary-trapped CO_2 and ϱ is the density of the single-phase CO_2. We set $\phi = \varphi(1-s)$ to instead work with the saturation s of capillary-trapped CO_2 and note that $\frac{\partial \phi}{\partial t} = -\varphi \frac{\partial s}{\partial t}$.

2.2.1 Boussinesq approximation

The first simplication to be made to the governing equations (2.3)-(2.7) is the Boussinesq approximation, which neglects variations in brine density ρ except in the buoyancy term. In (2.5) we set $\rho \approx \rho_{\text{ref}}$ for some constant reference brine density ρ_{ref} to obtain

$$\nabla \cdot \mathbf{u} = \varphi \frac{\partial s}{\partial t} + \frac{R}{\rho_{\text{ref}}} \tag{2.17}$$

$$= R\left(\frac{1}{\rho_{\text{ref}}} - \frac{1}{\varrho}\right) \tag{2.18}$$

and it follows that by expanding the left-hand side terms in (2.4) that

$$\varphi(1-s)\frac{\partial c}{\partial t} + \mathbf{u} \cdot \nabla c = \nabla \cdot (\mathsf{D} \cdot \nabla c) + R\left(1 - \frac{c}{\rho_{\text{ref}}}\right) \tag{2.19}$$

2.2.2 Streamfunction formulation

psi

2.2.3 Pressure formulation

p

2.2.4 Non-dimensionalisation & further approximation

Before discussing appropriate length, time and velocity scales it is helpful to first assume that the constitutive relations have certain scalings. Firstly we have from the aforementioned Boussinesq approximation $(\rho - \rho_{\text{ref}})g \sim g\Delta\rho$ for some density variation scale $\Delta\rho$. Likewise for a temperature scale $\Delta\theta$ we scale $\theta \sim \Delta\theta$. Secondly we expect the concentration of CO_2 in brine to be bounded from above by some equilibrium concentration c_{e} and hence $c \sim c_{\text{e}}$. To avoid confusion with the saturation s of capillary-trapped CO_2 we shall always refer to an equilibrium,

rather than saturation, concentration throughout this thesis. We make the scalings $\mu \sim \mu_{\text{ref}}$, $|\mathsf{K}| \sim K_{\text{ref}}$, $|\mathsf{G}| \sim G_{\text{ref}}$ and $|\mathsf{D}| \sim D_{\text{ref}}$ from appropriate reference values. Though the porosity is already a dimensionless quantity, it is instructive to rescale according to $\varphi \sim \varphi_{\text{ref}}$ and $\phi \sim \varphi_{\text{ref}}$ such that our rescaled porosities are of order unity. We also scale the reaction rate $R \sim \Delta R$. We do not yet make particular choices for the length, velocity and time scales which we shall denote by $\mathcal{L}$, $\mathcal{U}$, $\mathcal{T}$ to keep our non-dimensionalisation non-specific for now. A particular choice of $\mathcal{L}$, $\mathcal{U}$, $\mathcal{T}$ is informed by the details of a specific modelling scenario such as its boundary conditions or domain size and shape. Common choices for the length scale $\mathcal{L}$ include the domain length scale $\mathcal{L}_\Omega$ or diffusive length $D_{\text{ref}}/\mathcal{U}$. Similarly the velocity scale is often chosen as either the Darcy flow speed $K_{\text{ref}}\, g\Delta\rho/\mu_{\text{ref}}$ or diffusive speed $D_{\text{ref}}/\mathcal{L}$. Modelling scenarios considering the injection of fluid into the domain might also scale velocity with the injection speed u_{in} prescibed in the boundary conditions. A particular non-dimensionalisation can also be desirable for a numerical method to avoid numerical instabilities caused by excessively large or small terms. Finally the time scale can be constructed from the length and velocity scales as $\varphi_{\text{ref}}\mathcal{L}/\mathcal{U}$, or alternatively chose as either the reaction time scale $\varphi_{\text{ref}}c_{\text{e}}/\Delta R$ or porosity evolution time scale $\varphi_{\text{ref}}\varrho/\Delta R$. We use the velocity scale to scale the pressure as $p \sim \mu_{\text{ref}}\,\mathcal{U}\,\mathcal{L}/K_{\text{ref}}$ and define a non-dimensional reduced pressure

$$\hat{p} = \frac{K_{\text{ref}}}{\mu_{\text{ref}}\,\mathcal{U}\,\mathcal{L}}p + \frac{K_{\text{ref}}\rho_{\text{ref}}\,g}{\mu_{\text{ref}}\,\mathcal{U}}\mathbf{e}_g \cdot \hat{\mathbf{x}} \tag{2.20}$$

which in effect measures deviations from the hydrostatic pressure in brine at uniform reference density.

In the following we denote by a hat (e.g. $\mathbf{x} = \mathcal{L}\,\hat{\mathbf{x}}$) the non-dimensional and rescaled variables. After rescaling according to the definitions in Table Table 2.1, we obtain the non-dimensional governing equations

$$\widehat{\varphi}(1-s)\frac{\partial\hat{\theta}}{\partial\hat{t}} + \frac{\mathcal{U}\mathcal{T}}{\varphi_{\text{ref}}\,\mathcal{L}}\hat{\mathbf{u}}\cdot\nabla\hat{c} = \frac{G_{\text{ref}}\,\mathcal{T}}{\varphi_{\text{ref}}\,\mathcal{L}^2}\widehat{\nabla}\cdot(\widehat{\mathsf{G}}\cdot\widehat{\nabla}\hat{\theta}\,) \tag{2.21}$$

$$\widehat{\varphi}(1-s)\frac{\partial\hat{c}}{\partial\hat{t}} + \frac{\mathcal{U}\mathcal{T}}{\varphi_{\text{ref}}\,\mathcal{L}}\hat{\mathbf{u}}\cdot\nabla\hat{c} = \frac{D_{\text{ref}}\,\mathcal{T}}{\varphi_{\text{ref}}\,\mathcal{L}^2}\widehat{\nabla}\cdot(\widehat{\mathsf{D}}\cdot\widehat{\nabla}\hat{c}\,) + \frac{\mathcal{T}\,\Delta R}{\varphi_{\text{ref}}\,c_{\text{e}}}(1-\varepsilon\eta\hat{c}\,)\widehat{R} \tag{2.22}$$

$$\widehat{\nabla}\cdot\hat{\mathbf{u}} = -\varepsilon(1-\eta)\frac{\mathcal{L}\Delta R}{\mathcal{U}\,c_{\text{e}}}\widehat{R} \tag{2.23}$$

$$\hat{\mathbf{u}} = -\frac{\widehat{\mathsf{K}}}{\hat{\mu}}\cdot\left(\widehat{\nabla}\hat{p} - \frac{K_{\text{ref}}\rho_{\text{ref}}\,g}{\mu_{\text{ref}}\,\mathcal{U}}\hat{\rho}\,\mathbf{e}_g\right) \tag{2.24}$$

$$\widehat{\varphi}\frac{\partial s}{\partial\hat{t}} = -\varepsilon\frac{\mathcal{T}\,\Delta R}{\varphi_{\text{ref}}\,c_{\text{e}}}\widehat{R} \tag{2.25}$$

where

$$\eta = \frac{\varrho}{\rho_{\text{ref}}} < 1 \tag{2.26}$$

is the dimensionless ratio of the single-phase CO_2 density to the reference brine density and

$$\varepsilon = \frac{c_{\text{e}}}{\varrho} \ll 1 \tag{2.27}$$

is the dimensionless ratios of the brine's equilibrium concentration to the single-phase CO_2 density. Since single-phase CO_2 is known to be buoyant relative to brine, $\eta < 1$. Given that brine can only hold relatively small amounts of CO_2 in solution, $\varepsilon \ll 1$ is expected. To quantify the relative size of terms in (2.21)-(2.25), we also introduce the dimensionless parameter groupings

$$Ad = \frac{\mathcal{U}\mathcal{T}}{\varphi_{\text{ref}}\mathcal{L}} \tag{2.28}$$

$$Pe = \frac{\varphi_{\text{ref}}\,\mathcal{L}^2}{D_{\text{ref}}\,\mathcal{T}} \tag{2.29}$$

$$Ki = \frac{\mathcal{T}\,\Delta R}{\varphi_{\text{ref}}\,c_{\text{e}}} \tag{2.30}$$

$$Bu = \frac{K_{\text{ref}}\Delta\rho\, g}{\mu_{\text{ref}}\,\mathcal{U}} \tag{2.31}$$

and recall the definition of the Lewis number (ratio of thermal to solutal diffusivities)

$$Le = \frac{G_{\text{ref}}}{D_{\text{ref}}} \tag{2.32}$$

$$\tag{2.33}$$

Ad and Ki are sizes of the advection and reaction terms respectively. Pe is a Peclet-like number for the solutal diffusion term. Bu quantities the size of the buoyancy term and is in effect the ratio of the Darcy flow speed to the chosen velocity scale. We also denote by

$$Xl = \frac{\mathcal{L}_\Omega}{\mathcal{L}} \tag{2.34}$$

the non-dimensional domain's extent. In a specific modelling scenario with $\mathcal{L}$, $\mathcal{U}$, $\mathcal{T}$ chosen, the dimensionless groupings $\{Ad, Pe, Ki, Bu, X\}$ will map to familiar dimensionless numbers from fluid dynamics. We consider the $\varepsilon \to 0$ limit but retain ε in (2.25) to allow for spatio-temporal evolution of the saturation s, resulting in the further approximations

$$\widehat{\varphi}(1-s)\frac{\partial\hat{\theta}}{\partial\hat{t}} + Ad\,\hat{\mathbf{u}}\cdot\nabla\hat{c} = \frac{1}{LePe}\widehat{\nabla}\cdot(\widehat{\mathsf{G}}\cdot\widehat{\nabla}\hat{\theta}) \tag{2.35}$$

$$\widehat{\varphi}(1-s)\frac{\partial\hat{c}}{\partial\hat{t}} + Ad\,\hat{\mathbf{u}}\cdot\nabla\hat{c} \approx \frac{1}{Pe}\widehat{\nabla}\cdot(\widehat{\mathsf{D}}\cdot\widehat{\nabla}\hat{c}) + Ki\,\widehat{R} \tag{2.36}$$

$$\widehat{\nabla}\cdot\hat{\mathbf{u}} \approx 0 \tag{2.37}$$

$$\hat{\mathbf{u}} = -\frac{\widehat{\mathsf{K}}}{\hat{\mu}}\cdot(\widehat{\nabla}\hat{p} - Bu\,\hat{\rho}\,\mathbf{e}_g) \tag{2.38}$$

$$\widehat{\varphi}\frac{\partial s}{\partial\hat{t}} = -\varepsilon Ki\,\widehat{R} \tag{2.39}$$

Unless otherwise stated, we shall henceforth drop the hats from our dimensionless and rescaled variables. We summarise some common choices of $\mathcal{L}$, $\mathcal{U}$, $\mathcal{T}$ and their mappings in Table **??**, in which we define the solutal Rayleigh number (ratio of Darcy flow to solutal diffusion speeds, or equivalently ratio of solutal diffusion to Darcy flow time scales) as

$$Ra = \frac{\mathcal{L}_\Omega K_{\text{ref}}\, g\Delta\rho}{\mu_{\text{ref}} D_{\text{ref}}} \tag{2.40}$$

$$\tag{2.41}$$

(note that the related thermal Rayleigh number is $LeRa$) and the Damk"ohler number (ratio of the Darcy flow to reaction time scales) as

$$Da = \frac{\mathcal{L}_\Omega \mu_{\text{ref}} \Delta R}{K_{\text{ref}} g\Delta\rho\, c_{\text{e}}} \tag{2.42}$$

$$\tag{2.43}$$

Rb and Ra are readily identified as thermal and solutal Rayleigh numbers, respectively. Noting that G_{ref}/H and D_{ref}/H are diffusive speeds, we see that Rb and Ra are ratios of advective to diffusive speeds, or alternatively a ratio of diffusive to advective time scales, which in effect quantifies the strength of advective relative to solutal diffusive transport. We are primarily interested in the limits where advection is much quicker than diffusion, so $Rb \gg 1$ and $Ra \gg 1$.

Da is the Damköhler number for the dissolution reaction. Noting that the quantity $c_{\mathrm{e}}/\Delta R$ has units of time and may be interepreted as the dissolution time scale, we see that Da is a ratio of the advective to dissolution time scales. In the limit where dissolution is much quicker than advection, we have $Da \gg 1$.

Table 2.1: Non-dimensional and rescaled quantities

Quantity	Scale	Non-Dimensionalisation
$\mathbf{x}$	$\mathcal{L}$	$\hat{\mathbf{x}} = \mathbf{x}/\mathcal{L}$
θ	θ	$\hat{\theta} = (\theta - \theta_{\mathrm{ref}})/\Delta\theta$
c	c_{e}	$\hat{c} = c/c_{\mathrm{e}}$
$\mathbf{u}$	$\mathcal{U}$	$\hat{\mathbf{u}} = \mathbf{u}/\mathcal{U}$
t	$\mathcal{T}$	$\hat{t} = t/\mathcal{T}$
p	$\mu_{\mathrm{ref}}\,\mathcal{U}\,\mathcal{L}/K_{\mathrm{ref}}$	(2.20)
φ	φ_{ref}	$\widehat{\varphi} = \varphi/\varphi_{\mathrm{ref}}$
ϕ	φ_{ref}	$\hat{\phi} = \phi/\varphi_{\mathrm{ref}}$
ρ	$\Delta\rho$	$\hat{\rho} = (\rho - \rho_{\mathrm{ref}})/\Delta\rho$
μ	μ_{ref}	$\hat{\mu} = \mu/\mu_{\mathrm{ref}}$
K	K_{ref}	$\widehat{\mathsf{K}} = \mathsf{K}/K_{\mathrm{ref}}$
G	G_{ref}	$\widehat{\mathsf{G}} = \mathsf{G}/D_{\mathrm{ref}}$
D	D_{ref}	$\widehat{\mathsf{D}} = \mathsf{D}/D_{\mathrm{ref}}$
R	ΔR	$\widehat{R} = R/\Delta R$

Table 2.2: My Caption

$\mathcal{L}$	$\mathcal{U}$	$\mathcal{T}$	$\{Ad, Pe, Ki, Bu, Xl\}$
$\mathcal{L}_\Omega$	$K_{\mathrm{ref}}\,g\Delta\rho/\mu_{\mathrm{ref}}$	$\varphi_{\mathrm{ref}}\mathcal{L}/\mathcal{U}$	$\{1, Ra, Da, 1, 1\}$
$D_{\mathrm{ref}}/\mathcal{U}$	$K_{\mathrm{ref}}\,g\Delta\rho/\mu_{\mathrm{ref}}$	$\varphi_{\mathrm{ref}}\mathcal{L}/\mathcal{U}$	$\{1, 1, Da/Ra, 1, Ra\}$
$G_{\mathrm{ref}}/\mathcal{U}$	$K_{\mathrm{ref}}\,g\Delta\rho/\mu_{\mathrm{ref}}$	$\varphi_{\mathrm{ref}}\mathcal{L}/\mathcal{U}$	$\{1, Le, Da/(LeRa), 1, Ra/Le\}$
$\mathcal{L}_\Omega$	$D_{\mathrm{ref}}/\mathcal{L}$	$\varphi_{\mathrm{ref}}\mathcal{L}/\mathcal{U}$	$\{1, 1, RaDa, Ra, 1\}$
$\mathcal{L}_\Omega$	$D_{\mathrm{ref}}/\mathcal{L}$	$\varphi_{\mathrm{ref}}c_{\mathrm{e}}/\Delta R$	$\{1/(RaDa), RaDa, Ra, 1\}$
$D_{\mathrm{ref}}/\mathcal{U}$	$K_{\mathrm{ref}}\,g\Delta\rho/\mu_{\mathrm{ref}}$	$\varphi_{\mathrm{ref}}c_{\mathrm{e}}/\Delta R$	$\{Ra/Da, RaDa, 1, \ldots, \ldots\}$
$\sqrt{D_{\mathrm{ref}}\mathcal{T}/\varphi_{\mathrm{ref}}}$	$\varphi_{\mathrm{ref}}\mathcal{L}/\mathcal{T}$	$\varphi_{\mathrm{ref}}c_{\mathrm{e}}/\Delta R$	$\{1, 1, 1, \sqrt{Ra/Da}, \sqrt{RaDa}\}$

2.3 Diagnostic equations

2.3.1 Overview

We refer to the quantities $(\theta, c, \mathbf{u}, p, s)$ or $(\theta, c, \psi, \mathbf{u}, p, s)$ obtained by direct numerical solution of to (2.21)-(2.4) as primary quantities. Our postprocessing and modelling shall also make use of a number of so-called secondary quantities, derived from the primary quantities. For convenience and to avoid repetition in future chapters, we briefly define in this section some of these secondary quantities, and any methodology behind their computation. Some secondary quantities are amenable to being computed and saved 'on-the-fly' during the course of a numerical simulation, whereas others are better suited to being computed at the postprocess stage, for example using a third-party package from the scientific Python ecosystem or third-party visualization software such as ParaView [cite]. A summary of these quantities is tabulated in Table 2.3.

Symbol	Description	Method
$\min_{\mathbf{x}} \lvert\mathbf{u}\rvert$, $\max_{\mathbf{x}} \lvert\mathbf{u}\rvert$	minimum and maximum speed	(2.47), (2.48)
$\max_{\mathbf{x}}(q)$, $\min_{\mathbf{x}}(q)$ $\forall q \in \{\theta, c, s\}$	minimum and maximum of scalar fields	(2.44), (2.45)
$\langle q \rangle$ $\forall q \in \{\theta, c, s\}$	spatial-average of scalar fields	(2.44), (2.50)
m_C	capillary-trapped CO_2 mass	(2.53)
m_D	dissolved CO_2 mass	(2.54)
J	thermal flux	(2.60)
F	solutal flux	(2.61)
$N_{\uparrow\downarrow}$	number of plumes	
$u_{\uparrow\downarrow}$	average vertical velocity of plumes	
$w_{\uparrow\downarrow}$	average width of plumes	
k_P	plume wavenumber	

Table 2.3: A summary of secondary quantities

2.3.2 Global minima & maxima

Tracking the global minima and maxima of the temperature, concentration and velocity over the course of a numerical simulation is useful from both a physical and numerical perspective. Physically, we can identify certain behaviours, such as the onset of convection or attainment of an equilibrium state, from a timeseries plot of these scalar quantities. With the above discussion of numerical stability in mind, tracking minima and maxima of the temperature and concentration also serves as a diagnostic tool to detect spurious oscillations resulting in unphysical values outside of the physically-permitted range.

To demonstrate the computation of global minima and maxima, consider as examples the scalar-valued temperature $\theta^n = \sum_j \theta_j^n \xi_j$ and vector-valued velocity $\mathbf{u}^n = \sum_j u_j^n \xi_j$. In the

scalar-valued case, we first interpolate θ^n into the P_1 finite element function space, unless it is already the case that $\xi_j = \xi_j^{\mathrm{P}_1}$, to obtain $(\theta^{\mathrm{P}_1})^n = \sum_j (\theta^{\mathrm{P}_1})_j^n \xi_j^{\mathrm{P}_1}$. Then we have

$$\min_{\mathbf{x}}(\theta^n) = \min_j\{(\theta^{\mathrm{P}_1})_j^n\} \tag{2.44}$$

$$\max_{\mathbf{x}}(\theta^n) = \max_j\{(\theta^{\mathrm{P}_1})_j^n\} \tag{2.45}$$

In the case of a d-dimensional vector, we interpolate to a P_1^d vector space to obtain

$$(\mathbf{u}^{\mathrm{P}_1})^n = \sum_j \begin{pmatrix} (u_x^{\mathrm{P}_1})_j^n \\ (u_y^{\mathrm{P}_1})_j^n \\ \vdots \end{pmatrix} \xi_j^{\mathrm{P}_1} \tag{2.46}$$

Denoting by $|\cdot|$ the Euclidean vector norm, we have

$$\min_{\mathbf{x}} |\mathbf{u}^n| = \min_j \left\{ \sqrt{|(u_x^{\mathrm{P}_1})_j^n|^2 + |(u_y^{\mathrm{P}_1})_j^n|^+ \dots}^{1/2} \right\} \tag{2.47}$$

$$\max_{\mathbf{x}} |\mathbf{u}^n| = \max_j \left\{ \sqrt{|(u_x^{\mathrm{P}_1})_j^n|^2 + |(u_y^{\mathrm{P}_1})_j^n|^+ \dots}^{1/2} \right\} \tag{2.48}$$

In the case of a d-dimensional vector-valued quantity $\mathbf{u} = \sum_j U_j \xi_j$ with vector-valued basis functions $\{\xi_j\}$, the function must first be interpolated to a P_1^d vector space (if not already in such a space) and then the scalar-valued components extracted as $(u_x, u_y, \dots) = \sum_j (U_j^x, U_j^y, \dots)\xi_j^{\mathrm{P}_1}$. Using the conventional Euclidean norm, the global maximum is

$$\max_{\mathbf{x}} |\mathbf{u}| = \max_j \{ \sqrt{(U_j^x)^2 + (U_j^y)^2 + \dots}^{1/2} \} \tag{2.49}$$

and similarly for the global minimum. In our simulation we keep track not only of $\max_{\mathbf{x}} |\mathbf{u}|$, but also $\min_{\mathbf{x}} |\mathbf{u}|$, $\max_{\mathbf{x}}(c)$, $\min_{\mathbf{x}}(c)$, $\max_{\mathbf{x}}(s)$ and $\min_{\mathbf{x}}(s)$ as diagnostic quantities for both the providing insights into both the flow behaviour and validity of the numerical solution. Physically, extrema of the concentration field should obey the constraint $0 \leq c \leq 1$ at all times, and likewise for the saturation field $0 \leq s \leq s_r$. These constraints can be verified by a timeseries plot during postprocessing, or alternatively used during the course of the simulation to terminate a simulation producing spurious results.

2.3.3 Spatial averages

The spatial average of a quantity, over either a subdomain $\varpi \subset \Omega$ or spatial coordinate, will be relevant to a number of postprocessing calculations. We denote the operation of spatial averaging in 'bra-ket' notation as

$$\langle \theta \rangle_{\varpi}(t) = \frac{1}{\mathrm{vol}(\varpi)} \int_{\mathbf{x} \in \varpi} \theta(\mathbf{x}, t) \ \mathrm{d}\Omega \tag{2.50}$$

over a subdomain. In the case that Ω is a rectangule (spanned by x, y) or a cuboid (spanned by x, y, z), we define the horizontal average as

$$\langle \theta \rangle_{x_1 < x < x_2}(y, t) = \frac{1}{x_2 - x_1} \int_{x_1}^{x_2} \theta(x, y, t) \ \mathrm{d}x \tag{2.51}$$

$$\langle \theta \rangle_{x_1 < x < x_2, y_1 < y < y_2}(z, t) = \frac{1}{(x_2 - x_1)(y_2 - y_1)} \int_{y_1}^{y_2} \int_{x_1}^{x_2} \theta(x, y, z, t) \ \mathrm{d}x \mathrm{d}y \tag{2.52}$$

2.3.4 CO_2 mass

Our model of the dissolution of capillary-trapped CO_2 contains two contributions to the total amount of geologically-sequestered CO_2: the capillary-trapped mass

$$m_C(t) = \frac{1}{\varepsilon} \int_{\mathbf{x} \in \Omega} s(\mathbf{x}, t) \ \mathrm{d}\Omega \qquad = \frac{1}{\varepsilon} \int_0^L \int_{h(x,t)}^1 s(x, y, t) \ \mathrm{d}y \, \mathrm{d}x \tag{2.53}$$

\end{align}

and the dissolved mass

$$m_D(t) = \int_{\mathbf{x} \in \Omega} (1 - s(\mathbf{x}, t)) c(\mathbf{x}, t) \ \mathrm{d}\Omega \tag{2.54}$$

$$= \int_0^L \int_0^1 (1 - s(x, y, t)) c(x, y, t) \ \mathrm{d}y \, \mathrm{d}x \tag{2.55}$$

In a closed domain (i.e. no-flux boundary conditions everywhere on $\partial\Omega$), the total mass $m_C(t) + m_D(t)$ is a conserved quantity. The computation of domain integrals such as (2.53) and (2.54) is inexpensive and readily handled by the FEniCS library so the computation of m_C and m_D can be performed during the course of the numerical simulation.

2.3.5 Flux

We define the solutal and thermal fluxes, split into their advective and diffusive components, across a contour Γ respectively as

$$J_A(t;\Gamma) = \int_\Gamma \mathrm{d}\Gamma \; \mathbf{n} \cdot \mathbf{u}\, \theta \tag{2.56}$$

$$J_D(t;\Gamma) = \int_\Gamma \mathrm{d}\Gamma \; \frac{1}{Rb}\, \mathbf{n} \cdot (\mathsf{G} \cdot \nabla\theta) \tag{2.57}$$

$$F_A(t;\Gamma) = \int_\Gamma \mathrm{d}\Gamma \; \mathbf{n} \cdot \mathbf{u}\, c \tag{2.58}$$

$$F_D(t;\Gamma) = \int_\Gamma \mathrm{d}\Gamma \; \frac{1}{Ra}\, \mathbf{n} \cdot (\mathsf{D} \cdot \nabla c) \tag{2.59}$$

and the corresponding total fluxes

$$J(t;\Gamma) = J_A(t;\Gamma) + J_D(t;\Gamma) \tag{2.60}$$

$$F(t;\Gamma) = F_A(t;\Gamma) + F_D(t;\Gamma) \tag{2.61}$$

As with the above domain integrals for CO_2 mass, the FEniCS library computes contour integrals such as (2.56)-(2.59) with ease, provided that the contour Γ lies on the mesh's cell facets.

2.3.6 Events

As alluded to in our discussion of minima and maxima, we can identify from the timeseries $q(t)$ of some scalar-valued quantity certain specific times at which physically-interesting behaviours or regime changes occur. We shall refer any such specific times as an 'event'. For example, one possible definition for the onset of convection

$$t_{\mathrm{onset};\epsilon} = \operatorname*{argmin}_t \{ \max_{\mathbf{x}} |\mathbf{u}|(t) \;\; : \;\; \max_{\mathbf{x}} |\mathbf{u}|(t) > \epsilon \} \tag{2.62}$$

which corresponds to the time when the maximum speed first exceeds some small threshold value $0 < \epsilon \ll 1$. In a similar vein the shutdown of convection would be

$$t_{\text{shut};\epsilon} = \underset{t}{\text{argmax}}\{\max_{\mathbf{x}} |\mathbf{u}|(t) \ : \ \max_{\mathbf{x}} |\mathbf{u}|(t) > \epsilon\} \tag{2.63}$$

for when the maximum speed lass exceeds the threshold value. We defer the defition of further scenario-specific events to later chapters.

Event Description	Definition
onset of convection	$t_{\text{onset};\epsilon} = \underset{t}{\text{argmin}}\{\max_{\mathbf{x}} \lvert\mathbf{u}\rvert \ : \ \max_{\mathbf{x}} \lvert\mathbf{u}\rvert > \epsilon\}$
shutdown of convection	$t_{\text{shut};\epsilon} = \underset{t}{\text{argmax}}\{\max_{\mathbf{x}} \lvert\mathbf{u}\rvert \ : \ \max_{\mathbf{x}} \lvert\mathbf{u}\rvert > \epsilon\}$
plumes reach bottom boundary of domain	$t_{\text{bottom};\epsilon} = \underset{t}{\text{argmin}}\{\max_x c(x, y = 0, t) \ : \ \max_x c(x, 0, t) > \epsilon\}$
average concentration rises to value	$t_{\text{eq};c^*} = \underset{t}{\text{argmin}}\{\langle c\rangle_\Omega \ : \ \langle c\rangle_\Omega \geq c^*\}$
average saturation reaches drops to value	$t_{\text{diss};s*} = \underset{t}{\text{argmin}}\{\langle s\rangle_{\Omega^+} \ : \ \langle s\rangle_{\Omega^+} \leq s^*\}$

Table 2.4: Events

2.3.7 Plume statistics

We identify a set of solutal plumes as the contour at which concentration c is either equal to some specified value α

$$\Gamma_{P_c}(t;\alpha) = \{\mathbf{x} \in \Omega \ : \ c(\mathbf{x}, t) = \alpha\} \tag{2.64}$$

or a value at a fraction $0 \leq f \leq 1$ of its instantaneous range $[\min_{\mathbf{x}} c, \max_{\mathbf{x}} c]$

$$\Gamma_{P_c}(t;f) = \{\mathbf{x} \in \Omega \ : \ c(\mathbf{x}, t) = f \min_{\mathbf{x}} c(\mathbf{x}, t) + (1 - f) \max_{\mathbf{x}} c(\mathbf{x}, t)\} \tag{2.65}$$

Likewise we define thermal plumes $\Gamma_{P_\theta}(t;\alpha)$ and $\Gamma_{P_\theta}(t;f)$ with θ replacing c in the above contour definitions. Plume tips are then defined respectively as the set of cooordinates of the contour's local maxima and minima with respect to a coordinate x_P aligned with the direction of plume motion. In the case of plumes arising from fluid density contrasts, gravity is evidently the driving force for plume motion so $x_P = -\mathbf{e}_g \cdot \mathbf{x}$, which would simply be $x_P = y$ for a horizontal 2D domain and $x_P = z$ for a horizontal 3D domain. Hence for buoyant plumes we identify the ascending ($\uparrow$) and descending ($\downarrow$) plume tips at time t as

$$\{\mathbf{x}^{\uparrow j}(t)\}_{j=1}^{N^\uparrow(t)} = \{\mathbf{x} \ : \ \mathbf{x} \in \underset{x_P}{\text{locmax}}\ \Gamma_P(t)\} \tag{2.66}$$

$$\{\mathbf{x}^{\downarrow j}(t)\}_{j=1}^{N^\downarrow(t)} = \{\mathbf{x} \ : \ \mathbf{x} \in \underset{x_P}{\text{locmin}}\ \Gamma_P(t)\} \tag{2.67}$$

where $N^{\uparrow\downarrow}(t)$ is the total number of ascending and descending plumes. The trajectory of the jth plume tip is given by the sequence of coordinates $\{\mathbf{x}^{\uparrow\downarrow j}(t^n)\}_n$ for all times t^n in the time interval characterising the jth plume tip's lifetime (i.e. when that particular local maximum or minimum exists). The velocity of the jth plume tip is given by

$$u^{\uparrow\downarrow j}(t) = \frac{\mathrm{d}}{\mathrm{d}t}(x_P^{\uparrow\downarrow j}(t)) \tag{2.68}$$

so the average ascending and descending plume tip velocity is

$$u^{\uparrow\downarrow}(t) = \frac{1}{N_{\uparrow\downarrow}(t)} \sum_{j=1}^{N_{\uparrow\downarrow}} u^{\uparrow\downarrow j}(t) \tag{2.69}$$

The following calculations of plume width and wavenumber are generalizable to 3D but for clarity in our exposition we now consider a horizontal 2D domain. Given a fixed reference height y_{ref}, we compute the width of the jth plume by first defining its vertical extent as $y_{\text{ref}} \leq y \leq y^{\uparrow\downarrow j}$ or $y^{\uparrow\downarrow j} \leq y \leq y_{\text{ref}}$ in the case of ascending and descending plumes respectively. Next we set the height at which to measure to measure its width as $h_j = f \min(y^{\uparrow\downarrow j}, y_{\text{ref}}) + (1-f) \max(y^{\uparrow\downarrow j}, y_{\text{ref}})$ for some fraction $0 \leq f \leq 1$. By determining the coordinates $(x_-^{\uparrow\downarrow j}, h_j)$ and $(x_+^{\uparrow\downarrow j}, h_j)$ where the local plume contour intersects the horizontal line $y = h_j$, we deduce the plume width $w^{\uparrow\downarrow j} = x_+^{\uparrow\downarrow j} - x_-^{\uparrow\downarrow j}$. Similarly to (2.69) the average ascending and descending plume width is

$$w^{\uparrow\downarrow}(t) = \frac{1}{N_{\uparrow\downarrow}(t)} \sum_{j=1}^{N_{\uparrow\downarrow}} w_P^j(t) \tag{2.70}$$

The instantaneous solutal plume wavenumber at height y

$$k_{P_c}(y,t) = \frac{\int \; k' \mathbb{P}_c(k'; y, t) \mathrm{d}k'}{\int \mathbb{P}_c(k'; y, t) \mathrm{d}k} \tag{2.71}$$

is the expectation value of the power spectrum

$$\mathbb{P}_c(k'; y, t) = \lim_{T \to \infty} \frac{1}{T} \int |\hat{c}(k'; y, t)|^2 \; \mathrm{d}k' \tag{2.72}$$

of the Fourier-transformed concentration $\hat{c}(k'; y, t)$. Finally we denote by $\overline{k_{P_c}(y)}^{t,t'}$ the time-averaged wavenumber at height y over an interval $t \in [t, t']$. The thermal plume wavenumber $k_{P_\theta}(y,t)$ is likewise defined from the temperature θ.

2.3.8 Reaction front

Recall from **??** that we identify the reaction front height $h(x,t)$ as the curve partitioning the domain Ω into a reacted lower region $\Omega^-(t)$ where $s=0$ and an unreacted upper region $\Omega^+(t)$ having $0 < s \leq s_r$. Mathematically, this is defined as the countour

$$\Gamma_R(t) = \{(x,y) \in \Omega \ : \ s(x,y,t) = 0 \iff y = h(x,t)\} \tag{2.73}$$

though numerically we search using the `skimage` package's contour finder for $\Gamma_R(t;\epsilon_R) = \{(x,y) \in \Omega \ : \ s = \epsilon_R\}$ with some small tolerance $0 < \epsilon_R \ll 1$ to obtain $h(x,t;\epsilon_R)$. This method requires s to be represented by a `numpy` array data structure so computation is deferred to the postprocess stage. To quantify the nonlinear deformation of the initially horizontal reaction front as time progresses, we compute its normalized arc length

$$\mathrm{len}(h) = \frac{1}{L}\int_0^L \sqrt{1 + \left(\frac{\partial h}{\partial x}\right)^2}\ \mathrm{d}x \geq 1 \tag{2.74}$$

and standard deviation

$$\mathrm{sdev}(h) = \left(\frac{1}{L}\int_0^L (h - \langle h\rangle_x)^2\ \mathrm{d}x\right)^{1/2} \geq 0 \tag{2.75}$$

from the horizontally-averaged height $\langle h\rangle_x = \frac{1}{L}\int_0^L h(x,t)\ \mathrm{d}t$

2.3.9 Fluid front

Accompanying the concept of a reaction front, we have the fluid-fluid front, which for simplicity we abbreviate to fluid front, separating fluid rising into the unreacted upper region from the fluid at equilibrium concentration above it. As above, we define this mathematically as

$$\Gamma_F(t) = \{(x,y) \in \Omega \ : \ c(x,y,t) = 1 \iff y = h_F(x,t)\}$$

and implement it numerically with a small tolerance $0 < \epsilon_F \ll 1$ such that `skimage` searches for the contour $\Gamma_F(t;\varepsilon_F) = \{(x_i,y_i) \in \Omega_{N_x,N_y} \ : \ c = 1 - \epsilon_F\}$

2.4 Numerical methods

2.4.1 Discretization in time

To solve (2.21)-(2.25) numerically, we use the finite element method in space with finite differences in time taken before discretization in space. Denoting by t^n, Δt^n the time and timestep at the nth step of the numerical integration, such that $\theta(\mathbf{x}, t^n) \approx \theta^n(\mathbf{x})$ and the initial time is $t^0 = 0$, the discretized system at the nth step is

$$\mathcal{D}_\phi(\phi)\frac{\theta^{n+1} - \theta^n}{\Delta t^n} + \mathcal{D}_{A,\theta}(\mathbf{u} \cdot \nabla\theta) = \frac{1}{Rb}\nabla \cdot \mathcal{D}_{D,\theta}(\mathsf{G} \cdot \nabla\theta) \tag{2.76}$$

$$\mathcal{D}_\phi(\phi)\frac{c^{n+1} - c^n}{\Delta t^n} + \mathcal{D}_{A,c}(\mathbf{u} \cdot \nabla c) = \frac{1}{Ra}\nabla \cdot \mathcal{D}_{D,c}(\mathsf{D} \cdot \nabla c) + Da\,\mathcal{D}_{R,c}(R) \tag{2.77}$$

$$\nabla \cdot \mathbf{u}^n = 0 \tag{2.78}$$

$$\mathbf{u}^n = -\frac{\mathsf{K}^n}{\mu^n}(\nabla p^n + \rho^n \mathbf{e}_g) \tag{2.79}$$

$$\varphi\frac{s^{n+1} - s^n}{\Delta t^n} = -\varepsilon Da \mathcal{D}_{R,s}(R) \tag{2.80}$$

and is to be solved for θ^{n+1} c^{n+1}, $\mathbf{u}^n$, p^n and s^{n+1}. $\mathcal{D}$ denotes a finite difference operator acting on a time-dependent quantity or expression to produce a discretization in time. For example, a second-order Adams-Bashforth operator $\mathcal{D} = \text{AB}_2$ produces

$$\text{AB}_2(\theta) = \frac{3}{2}\theta^n - \frac{1}{2}\theta^{n-1} \tag{2.81}$$

which is explicit with respect to θ, and the backward Euler operator $\mathcal{D} = \text{BE}$ produces

$$\text{BE}(\theta) = \theta^{n+1} \tag{2.82}$$

which is implicit with respect to θ since θ^{n+1} is being solved for. A general finite difference operator can be defined as producing a linear superposition of future, present and previous values of the time-dependent quantity of expression

$$\mathcal{D}(\theta) = \sum_{j \leq 1} \alpha_j \theta^{n+j} \tag{2.83}$$

according to its set of constant coefficients $\{\alpha_j\}$. Note that $\alpha_1 = 0$ produces an explicit discretization. When discretizing an expression, such as the advection $a(\mathbf{u}, \theta) = \mathbf{u} \cdot \nabla\theta$, it possible to discretize expression-wise as

$$\mathcal{D}(a(\mathbf{u}, \theta)) = \sum_{j \leq 1} \alpha_j a(\mathbf{u}^{n+j}, \theta^{n+j}) \tag{2.84}$$

or if $\mathcal{D}$ is specified as the tuple $(\mathcal{D}_{\mathbf{u}}, \mathcal{D}_\theta)$ then we discretize argument-wise as

$$(\mathcal{D}_{\mathbf{u}}, \mathcal{D}_\theta)(a(\mathbf{u}, \theta)) = a(\mathcal{D}_{\mathbf{u}}(\mathbf{u}), \mathcal{D}_\theta(\theta)) \tag{2.85}$$

In our programming of a numerical solution we avoid hard-coding any particular finite difference discretizations, instead allowing each $\mathcal{D}$ to be specified as an input parameter to the numerical simulation program. A more detailed discussion of our implementation is deferred to **??**.

To satisfy stability criteria with respect to both advection and reaction, the timestep Δt^n must be computed adaptively at each step by choosing whicever is smallest between the Courant–Friedrichs–Lewy (CFL) and reaction timesteps Δt^n_{CFLR}, Δt^n_R. We define such as timestep as

$$\Delta t^n_{\text{CFLR}} = \min\{C_{\text{CFL}} \Delta t^n_{\text{CFL}}, C_R \Delta t^n_R\} \tag{2.86}$$

where

$$\Delta t^n_{\text{CFL}} = \min_{\mathbf{x}} \left(\frac{\ell(\mathbf{x})}{|\mathbf{u}^n|} \right) \tag{2.87}$$

$$\Delta t^n_R = \frac{1}{\max_{\mathbf{x}}(Da|R^n|)} \tag{2.88}$$

where $0 < C_{\text{CFL}}, C_R \leq 1$ are safety factors and $\ell(\mathbf{x})$ is a local measure of mesh cell size (e.g. diameter, circumradius or minimum edge length). Note that in the case of a uniform mesh we have the simplication $\ell(\mathbf{x}) = \ell$ and hence

$$\Delta t^n_{\text{CFL}} = \frac{\ell}{\max_{\mathbf{x}} |\mathbf{u}^n|} \tag{2.89}$$

Finally, for practical purposes we also restrict the timestep to a desired range such that $\Delta t_{\min} \leq \Delta^n \leq \Delta t_{\max}$ by implementing

$$\Delta t^n = \min\{\Delta t_{\max}, \max\{\Delta t_{\min}, \Delta t^n_{\text{CFLR}}\}\} \tag{2.90}$$

Rather than solving the entire system (2.76)-(2.80) at once as a nonlinear, mixed finite element problem for θ^{n+1}, c^{n+1}, $\mathbf{u}^n$, p^n and s^{n+1}, we solve each subsystem in turn as a linear problem. In particular, we note that (2.78) and (2.79) are level in time and hence the flow subsystem can be solved independently for $(\mathbf{u}^n, p^n)$, given K^n, μ^n and ρ^n which are in turn given in terms of θ^n, c^n and s^n. In the streamfunction formulation, (2.78) and (2.79) are replaced by

$$\nabla \cdot \left(\frac{\mu^n \left(\mathsf{K}^n\right)^{\mathsf{T}} \cdot \nabla \psi^n}{\det(\mathsf{K}^n)} \right) = -\frac{\partial (e_g^y \rho^n)}{\partial x} + \frac{\partial (e_g^x \rho^n)}{\partial y} \tag{2.91}$$

$$\mathbf{u}^n = \begin{pmatrix} \frac{\partial \psi^n}{\partial y} \\ -\frac{\partial \psi^n}{\partial x} \end{pmatrix} \tag{2.92}$$

Once $\mathbf{u}^n$ is known, the transport equations (2.76) and (2.77) can be solved in turn, provided that their discretizations are explicit with respect to $\mathbf{u}$ and s. The final problem to solve is (2.80), and since θ^{n+1} and c^{n+1} are known from solving (2.76) and (2.77) in advance, its discretization can be explicit with respect to θ and c. However if $R(s, c, \theta)$ is nonlinear in s then to obtain a linear problem the discretization must be explicit with respect to s. We list the complete numerical integration algorithm in Algorithm **??**. More complicated numerical integration algorithms, such as predictor-corrector methods, are also possible for this system but for simplicity we do not pursue such approaches.

3 Numerical software

```python
import lucifex
import crocodil
print('hello world')
```

```
hello world
```

```python
from lucifex.fdm import FiniteDifference

print(FiniteDifference)
```

```
<class 'lucifex.fdm.finite_difference.FiniteDifference'>
```

```python
print(FiniteDifference)
```

```
<class 'lucifex.fdm.finite_difference.FiniteDifference'>
```

References